

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Third Semester of B. Tech (IT) Examination****April-May 2015****IT217 Digital Logic & Design****Date: 27.04.2015, Monday****Time: 10.00 a.m. To 01.00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the questions below.****[07]**

- a. Which of the following code is known as 8421 code?
 - 1) Gray Code
 - 2) XS-3 Code
 - 3) BCD code
- b. A bubbled NAND gate is equivalent to a _____ gate.
- c. Which are the most suitable gates to check whether the number of 1s in the digital word is even or odd?
- d. Define Propagation Delay.
- e. What is the special advantage of CMOS logic family?
- f. Convert 100100100111.010010 to the octal.
- g. Difference between Standard form and Canonical Form.

Q – 2.a 1. Find 9's complement of $(578)_9$ **[04]**

2. Convert decimal number 100 to Base3, Base 4.

Q – 2.b Answer any two questions.**[10]**

1. State and prove the De Morgan's law using truth table.
2. Design full adder circuit.
3. Reduce the following expressions
 - i. $F = A[B + C'(AB+AC')]'$
 - ii. $F = (B + BC)(B + B'C)(B+D)$

Q - 3 Answer the following.

- a. Obtain minimal expression for the Boolean expression [07]
 $F = \sum m(1,2,3,5,6,7,8,9,12,13,15)$ using the tabulation method.
- b. 1. Implement a full subtractor circuit with a Decoder and two OR gate. [04]
 2. Express the Boolean function $F = XY + X'Z$ in a product of maxterm. [03]

OR

1. Explain a two bit magnitude comparator with the diagram [04]
 2. Implement $F(a,b,c) = \sum m(0,1,3,5,7)$ with the help of multiplexer [03]

SECTION – II**Q - 4 Answer the question below. [07]**

- a. Which flip-flop is considered as Universal flip-flop?
- b. For the Positive Edge D flip flop the Input condition is: Clock is changing from Low to high, Input at D is 1 then what should be the output?
 1) Memory state
 2) Toggling
 3) 0
 4) 1
- c. What are called as don't care conditions?
- d. What is master- slave flip-flop?
- e. What is EEPROM?
- f. Define State Diagram.
- g. List out the applications of flip-flops.

Q – 5.a Reduce the following function using K-Map and implement it with NAND Gate, [05]

$$F(a,b,c,d) = \sum m(0,1,2,3,5,7,8,9,10,12,13)$$

OR**Q – 5.a Design the combinational circuit that accept 2 bit binary number and generates an output binary numbers equal to the square of input number. [05]****Q – 5.b Give the comparison between synchronous and asynchronous counter. [04]**

- Q – 5.c** Reduce the number of state in the following state stable and tabulate reduced state table [05]
along with state diagram.

Present state	Next State		Output	
	X = 0	X = 1	X = 0	X = 1
a	f	b	0	0
b	d	c	0	0
c	f	e	0	0
d	g	a	1	0
e	d	c	0	0
f	f	b	1	1
g	g	h	0	1
h	g	a	1	0

OR

- Q – 5.c** Explain 4 bit bidirectional shift register with diagram. [05]
- Q – 6.a** Why flip flop is called a single bit register? [02]
- Q – 6.b** Design Mod-6 Counter using T- Flip flop. [07]
- Q – 6.c** What is race around condition? How can we remove it? [03]
- Q – 6.d** Make characteristic table and excitation table for JK and D flip flop. [02]

OR

- Q – 6.a** Write a short note on Johnson Counter which can generate 8 timing sequence. [06]
- Q – 6.b** Define the following term with respect to sequential circuit. [03]
1) State 2) Characteristic Table 3) Excitation Table
- Q – 6.c** Write short note on Memory Unit. [05]
